STORMTRACK

- **Problem Statement ID – 26078**
- **Problem Statement Title-**

  AI-Driven Spatio-Temporal Tracking of Extreme WeatherAnomalies in Medium-Range Forecasts
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal) - Baby ko Database Pasand Hai**

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PROPOSED SOLUTION

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PROBLEM AT HAND

Fragmented signal

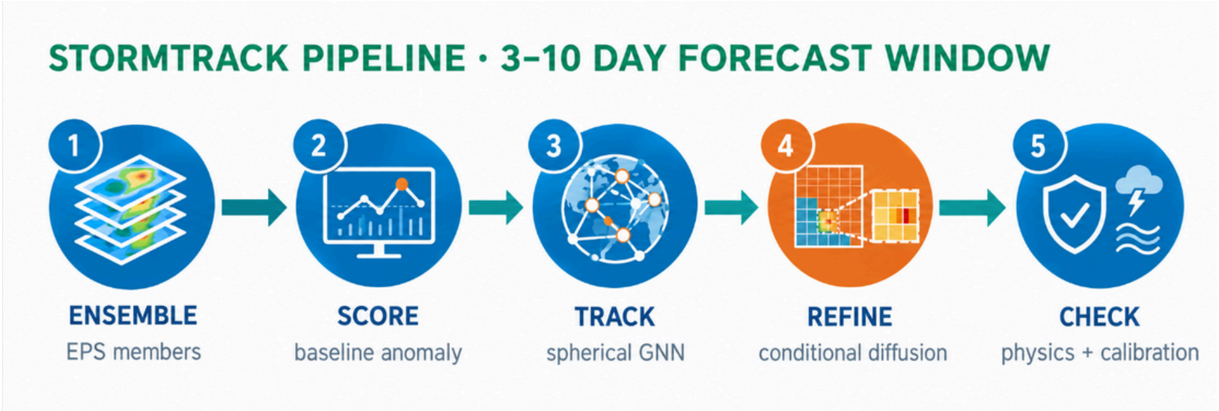
Extreme pockets hide across forecast times and ensemble members.

Manual triage

Reviewing fields and regions takes time during active events.

Broad alerts

Large warning footprints can overwhelm local users.



12 km forecast crop → proposed 5 km scenario set

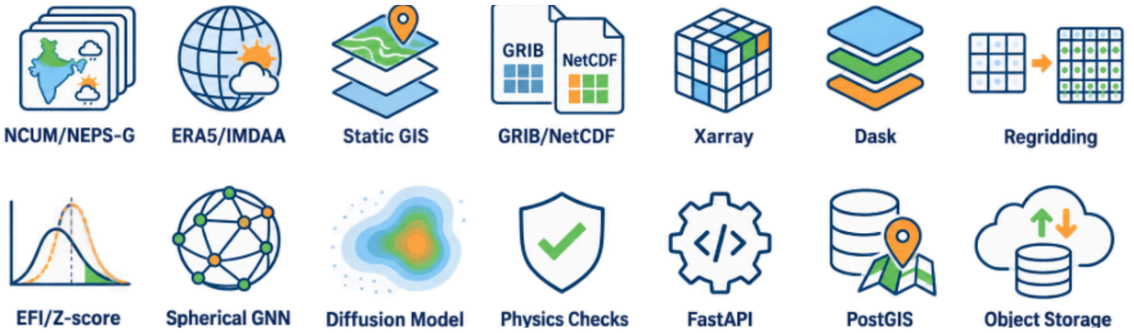
Localized map layer with hazard, lead time and confidence range

Design target: sharper local guidance with uncertainty visible.

DECISION OUTPUT

- **Anomaly footprint** — Where risk clusters
- **Day 3–10 timeline** — When the signal evolves
- **Risk + confidence** — How certain members are

TOOLS USED



VIDEO SOLUTION

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TECHNICAL APPROACH

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1. Data Sources & Input

NCUM/NEPS-G forecast ensembles + ERA5/IMDAA climatology + static GIS layers

2. Data Ingestion & Preprocessing

GRIB/NetCDF → Xarray/Dask → regridding → feature tensors

3. Anomaly Intelligence Layer

climatology comparison → EFI/z-score → extreme anomaly masks

4. Spatio-Temporal Tracking Layer

spherical GNN on icosahedral mesh → event detection → trajectory

5. Local Downscaling Layer

conditional diffusion model → 12 km coarse forecast to 5 km risk field

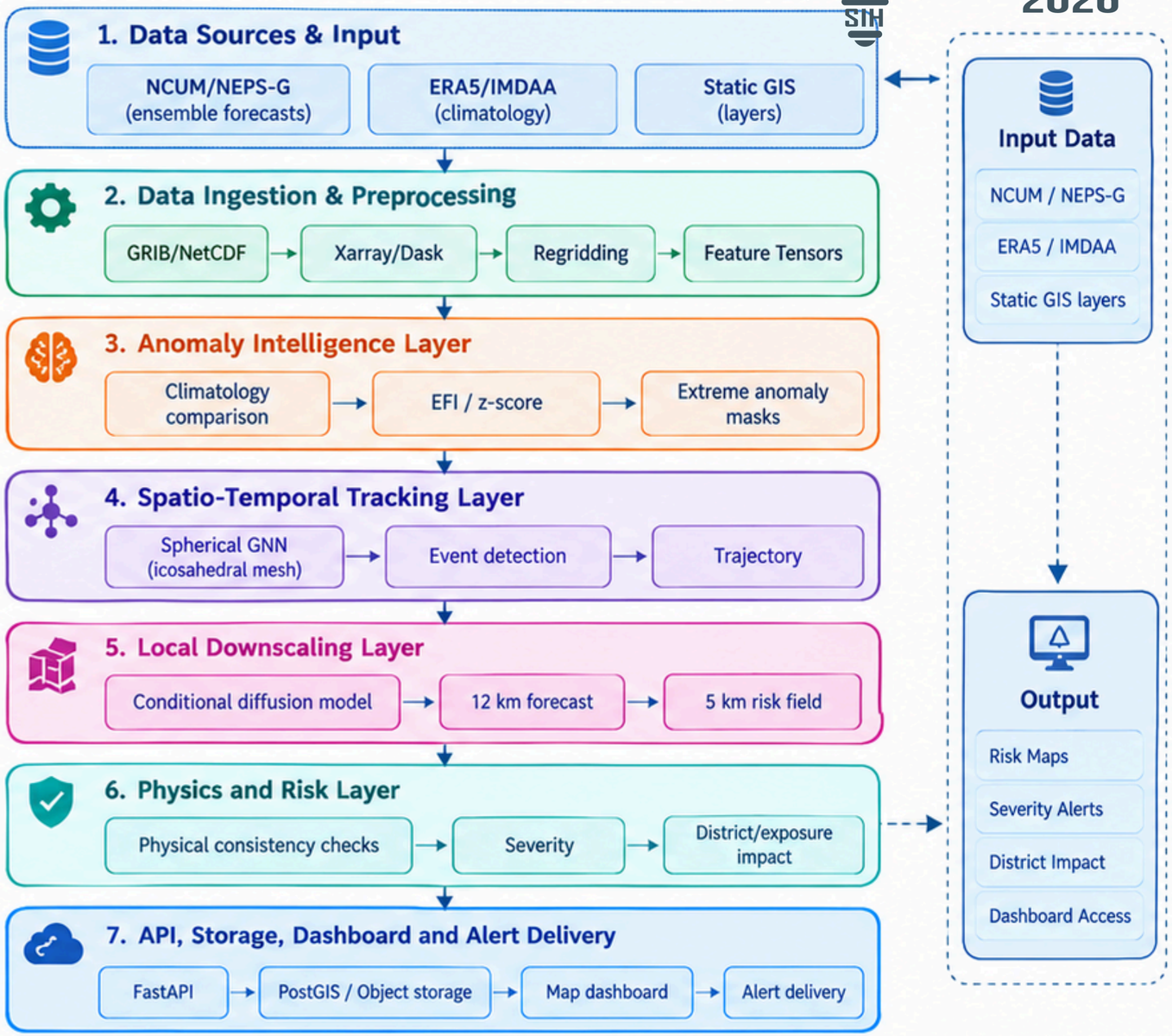
6. Physics and Risk Layer

physical consistency checks → severity → district/exposure impact

7. API, Storage, Dashboard and Alert Delivery

FastAPI → PostGIS/object storage → map dashboard → alert delivery

Python Xarray Dask netCDF PostGIS FastAPI React GeoServer

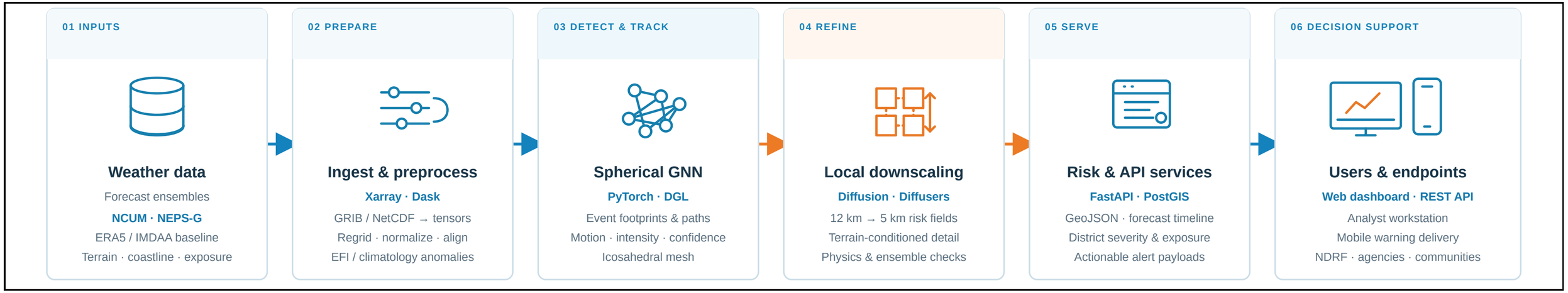


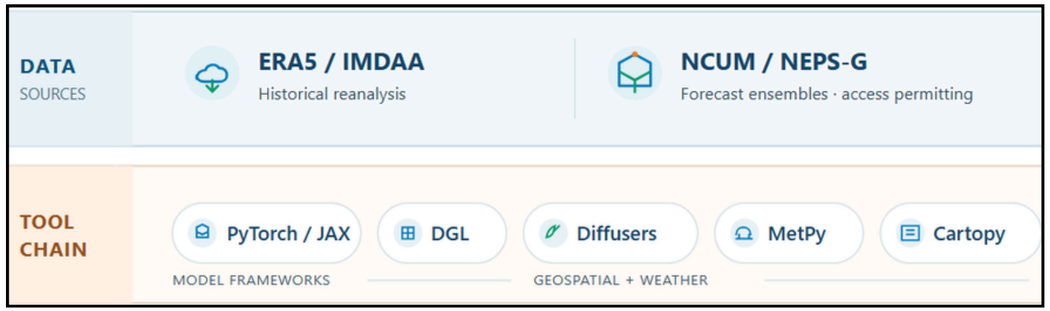
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FEASIBILITY AND VIABILITY

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SYSTEM ARCHITECTURE





EVAL METRICS

FEASIBILITY CHECKS

Start with a few variables and 1–2 documented events. Use archived or sample data if live EPS access is unavailable.

Risks: data rights, event imbalance, member consistency, calibration and compute.

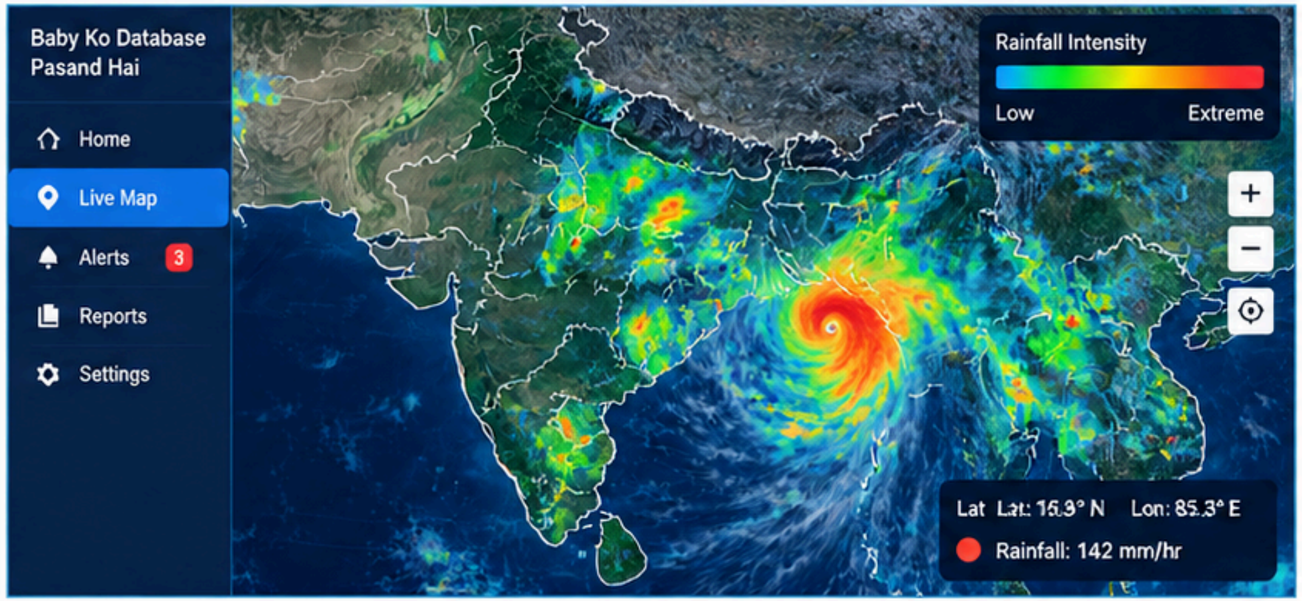
Detection precision / recall	Track Error	Downscaling RMSE / CRPS
Extreme Tail Preservation	Footprint IoU	Confidence Calibration + Latency

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IMPACT AND BENEFITS

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ILLUSTRATIVE DASHBOARD



SEVERE RAINFALL WATCH

LOCATION	Bay of Bengal (Eastern Coast of India)
LEAD TIME	Day 5 – 7
FOOTPRINT	Centroid + proposed 5 km risk layer
CONFIDENCE	High (based on ensemble + AI analysis)
ACTION	Issue early alerts, enable disaster response, and guide evacuation planning

KEY IMPACTS AND BENEFITS

DISASTER RESPONSE

Helps authorities prioritize reconnaissance, logistics and rescue operations in affected areas.

AGRICULTURE

Protects crops by providing early warnings for extreme rainfall, heat stress and flooding, reducing crop loss.

URBAN SYSTEMS

Assists in smart drainage planning, reduces urban flooding, and enables better asset and infrastructure checks.

Overall, the project enables faster, data-driven decisions, reduces loss of life and property, and builds a more resilient and prepared community.

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RESEARCH AND REFERENCES

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References

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   Copernicus Climate Data Store
   https://cds.climate.copernicus.eu/datasets/reanalysis-era5-complete

2. **IMDAA reanalysis**
   NCMRWF · 12 km regional reanalysis
   https://nwp.ncmrwf.gov.in/reanalysis

3. **GraphCast**
   Lam et al. · Science (2023)
   https://deepmind.google/research/publications/22598/

4. **Extreme Forecast Index**
   ECMWF Forecast User Guide
   https://confluence.ecmwf.int/spaces/FUG/pages/673551296/Section+8.1.9.2+Extreme+Forecast+Index+-+EFI

5. **Diffusion downscaling**
   Dai et al. · JGR: Machine Learning and Computation (2026)
   https://doi.org/10.1029/2026JH001347

6. **Probabilistic forecasting**
   Price et al. · Nature (2025), GenCast
   https://doi.org/10.1038/s41586-024-08252-9

Comparison with Existing Systems

Feature	Proposed System	SAARTHI (Ours)	AlmaConnect	Gradway	FiberVite
4D ensemble input	✓	✓	✗	✗	✗
Spherical GNN anomaly tracking	✓	✗	✗	✗	✗
EFI-based anomaly detection	✓	✓	✗	✗	✗
3–10 day anomaly trajectory	✓	✓	✓	✗	✗
5 km statistical downscaling	✓	✗	✗	✗	✗
Amplitude-preserving extremes	✓	✗	✗	✗	✗
Physics-informed constraints	✓	✗	✗	✗	✗
Probabilistic output	✓	✓	✓	✗	✗
Hyper-local risk zone	✓	✗	✗	✗	✗
Automated alert API	✓	✗	✗	✗	✗

✓ = Available / Supports ✗ = Not Available – = Partial / Limited

Research Workflow

Data Collection (ERA5 + IMDAA) → Preprocessing (Xarray + Dask) → GNN Tracking (12 km → icosahedral mesh) → Diffusion Downscaling (12 km → 5 km) → Visualization & Alerts (Dashboard + API)